Занятие VII

Supporting Vector Machines

Повторение

1. Метрики качества классификации

- Доля правильных ответов (accuracy):

$$\frac{1}{\ell}\sum_{i=1}^{\ell}[a(x_i) = y_i]$$

	$y = 1$	$y = -1$
$a(x) = 1$	True Positive (TP)	False Positive (FP)
$a(x) = -1$	False negative (FN)	True Negative (TN)

Таблица 1. Матрица ошибок

- Можно ли доверять классификатору при $a(x)$

$$\text{precision}(a, X) = \frac{TP}{TP + FP}$$

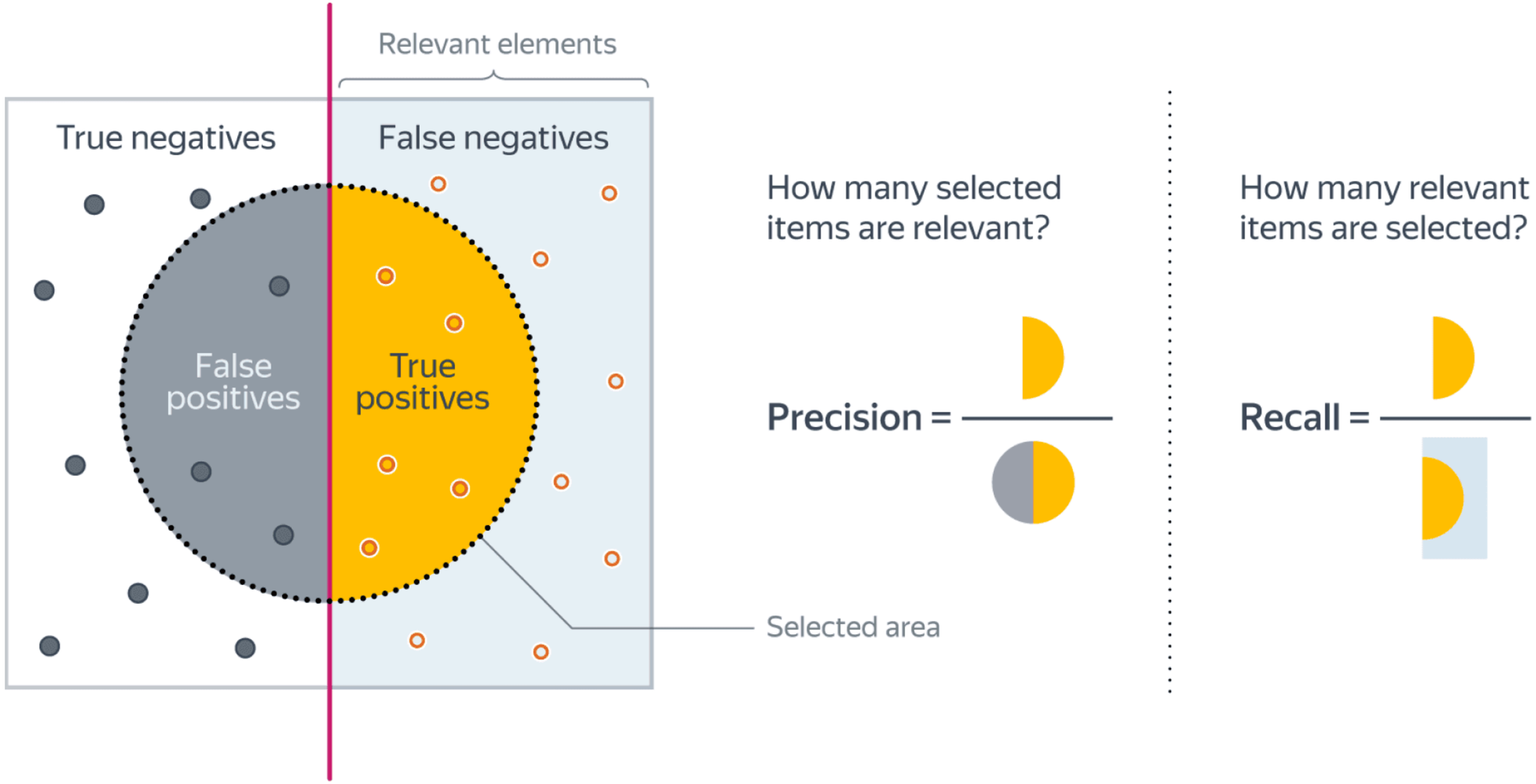
- Как много положительных объектов находит классификатор?

$$\text{recall}(a, X) = \frac{TP}{TP + FN}$$

1. Метрики качества классификации

		Predicted Class: Positive	Predicted Class: Negative	
Actual Class	Positive	True Positive (TP)	False Negative (FN) Type II Error	Recall $\frac{TP}{(TP+FN)}$
	Negative	False Positive (FP) Type I Error	True Negative (TN)	
		Precision $\frac{TP}{(TP+FP)}$		**Accuracy** $\frac{TP+TN}{(TP+TN+FP+FN)}$

1. Метрики качества классификации

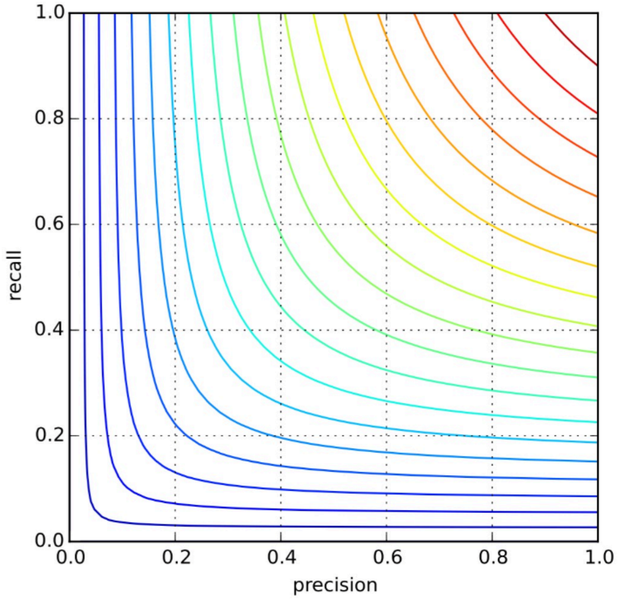


1. Метрики качества классификации

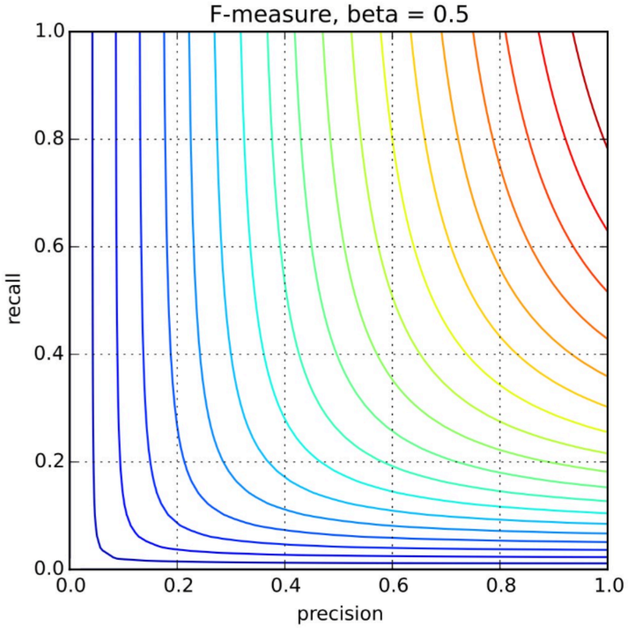
$$F_\beta = (1+\beta^2)\frac{\text{precision} * \text{recall}}{\beta^2 * \text{precision} + \text{recall}}$$

F1-мера

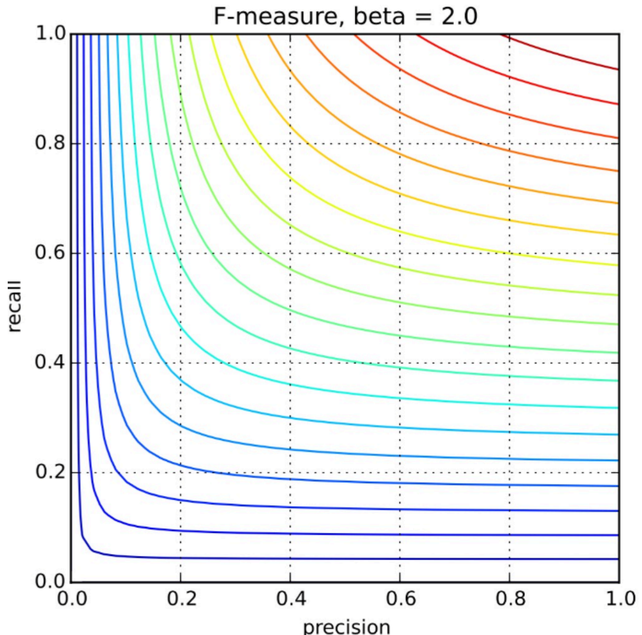
$$F = \frac{2 * \text{precision} * \text{recall}}{\text{precision} + \text{recall}}$$



- $\beta = 0.5$
- Важнее точность



- $\beta = 2$
- Важнее полнота



1. Метрики качества классификации

- Высокий порог:
  - Мало объектов относим к +1
  - Точность выше
  - Полнота ниже
- Низкий порог:
  - Много объектов относим к +1
  - Точность ниже
  - Полнота выше

- Пример: кредитный скоринг
- $b(x)$ — оценка вероятности возврата кредита
- $a(x) = [b(x) > 0.5]$
- $\text{precision} = 0.1, \text{recall} = 0.7$
- В чем дело — в пороге или в алгоритме?

True Negative | True Positive

False Negative | False Positive

a(x_i) = -1 | a(x_i) = 1

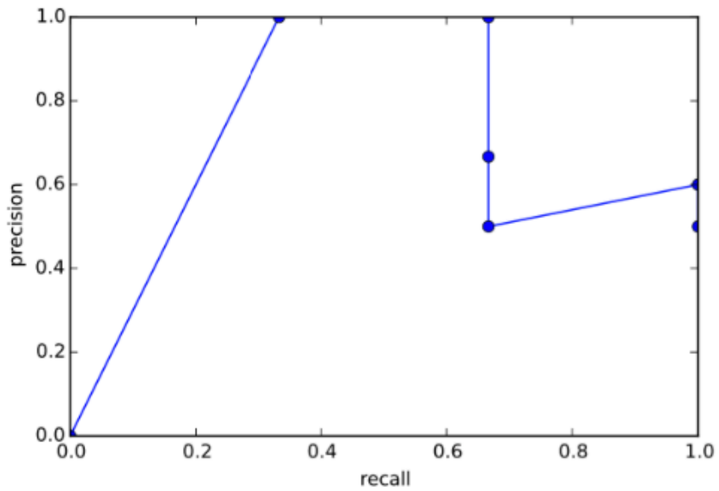
y_i	-1	-1	+1	-1	-1	-1	+1	+1	-1	+1
b(x_i)	0.01	0.09	0.12	0.15	0.29	0.4	0.48	0.6	0.83	0.9

Recall = TP / (TP + FN) = 0.5

Precision = TP / (TP + FP) = 2/3

1. Метрики качества классификации

- Кривая точности-полноты
- Ось X — полнота
- Ось Y — точность
- Точки — значения точности и полноты при последовательных порогах



$b(x)$	0.14	0.23	0.39	0.52	0.73	0.90
y	0	1	0	0	1	1
	P =0.6; R = 1	P = 1/2; R = 2/3	P = 2/3; R = 2/3	P = 1; R = 2/3	P = 1; R = 1/3	

1. Метрики качества классификации

- Receiver Operating Characteristic
- Ось X — False Positive Rate

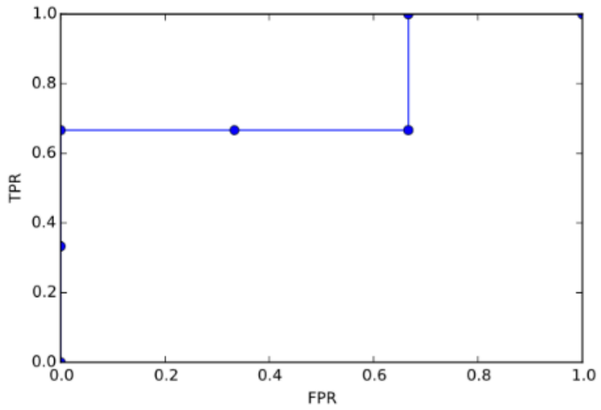
$$FPR = \frac{FP}{\boxed{FP + TN}}$$

Число отрицательных объектов

- Ось Y — True Positive Rate

$$TPR = \frac{TP}{\boxed{TP + FN}}$$

Число положительных объектов



$b(x)$	0.14	0.23	0.39	0.52	0.73	0.90
y	0	1	0	0	1	1

TPR=1
FPR=1

TPR=1
FPR=2/3

TPR=2/3
FPR=2/3

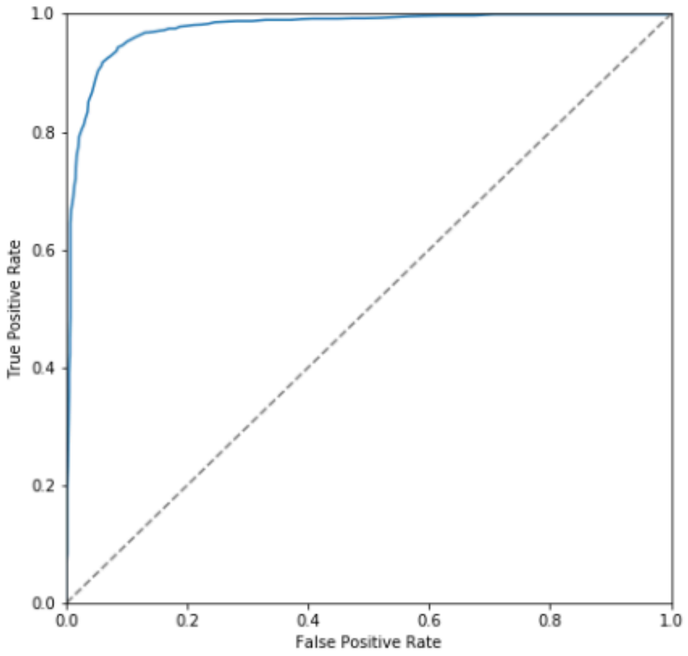
TPR=2/3
FPR=1/3

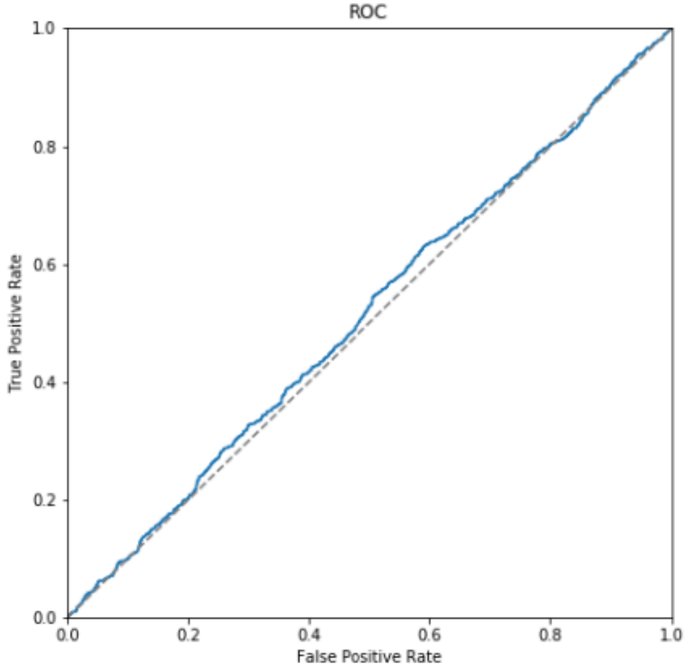
TPR=2/3
FPR=0

TPR=1/3
FPR=0

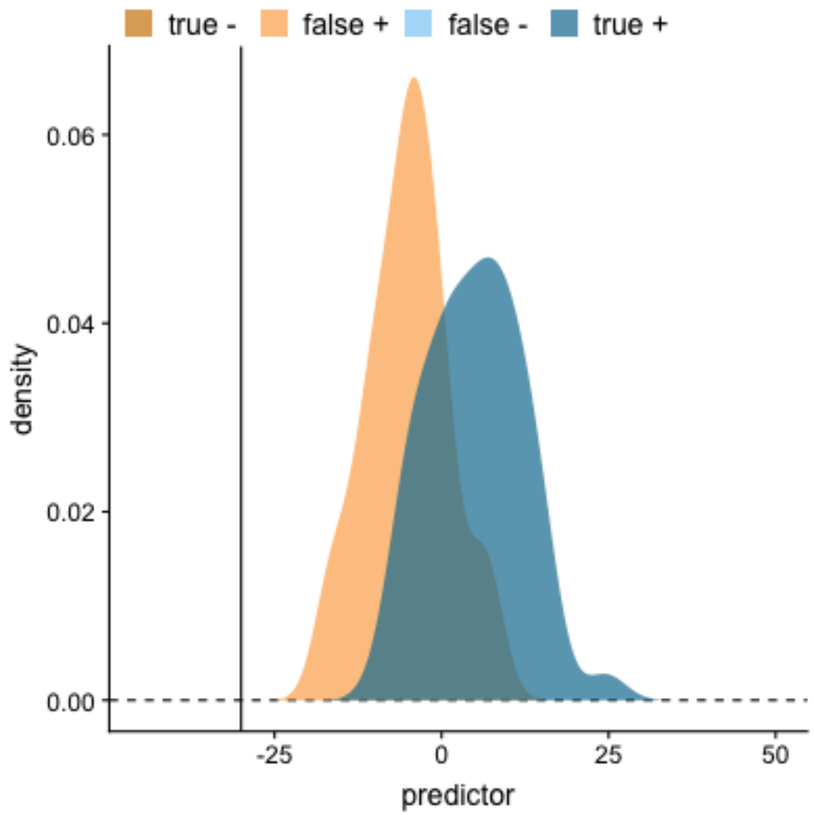
1. Метрики качества классификации

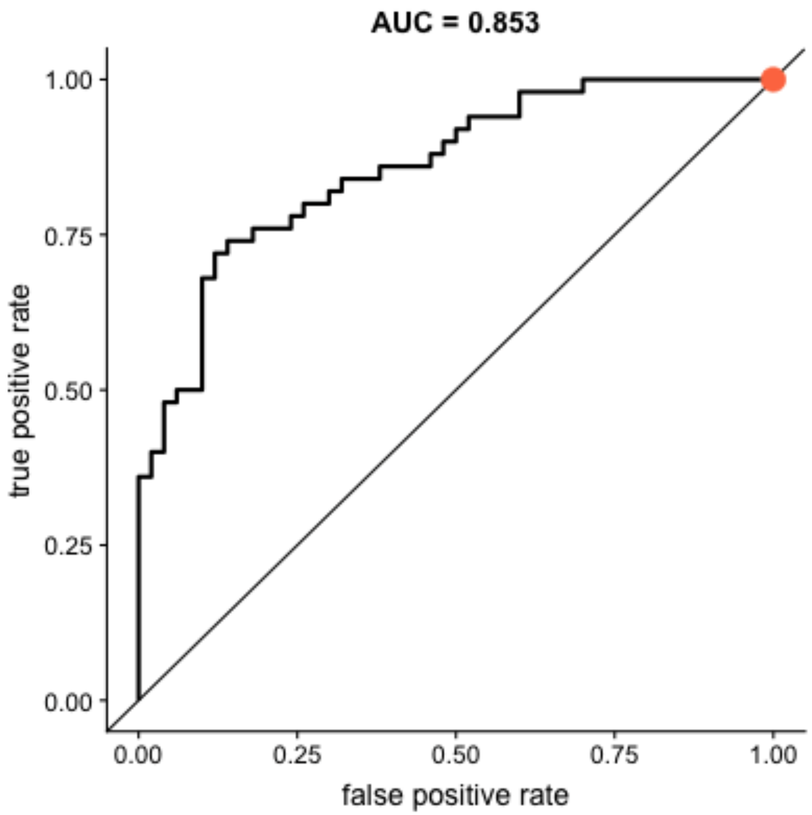
- Левая точка: (0, 0)
- Правая точка: $(1, 1)$
- Для идеального классификатора проходит через (0, 1)
- AUC-ROC — площадь под ROC-кривой





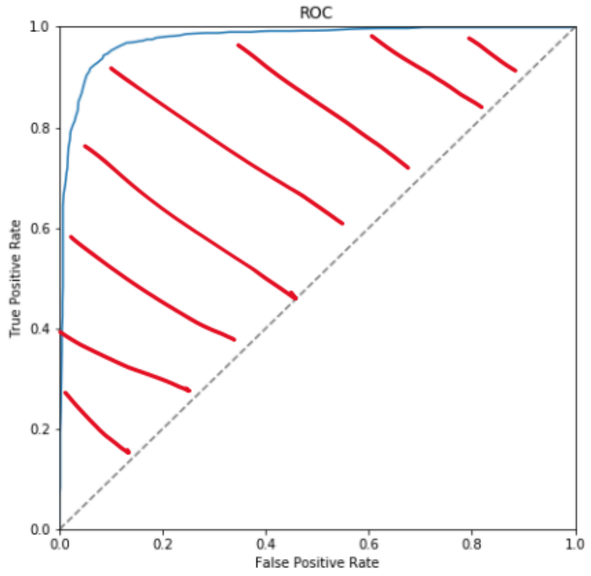
1. Метрики качества классификации

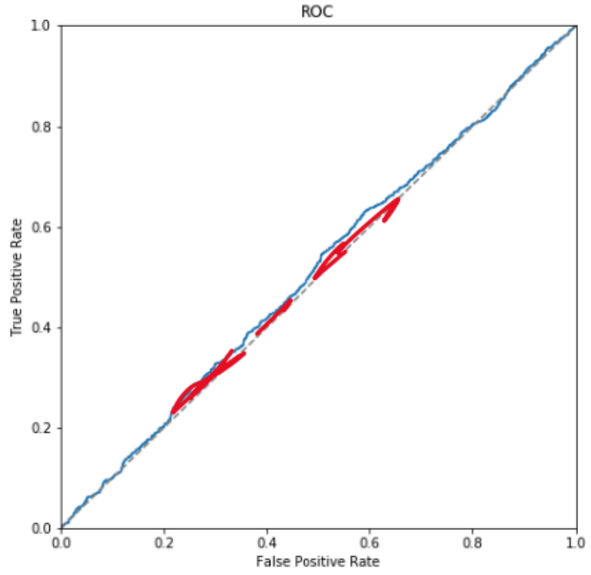




1. Метрики качества классификации

$$\text{Gini} = 2 * (\text{AUC}-\text{ROC} - 0.5)$$





ЛогРег:
простое объяснение

2. Логистическая регрессия

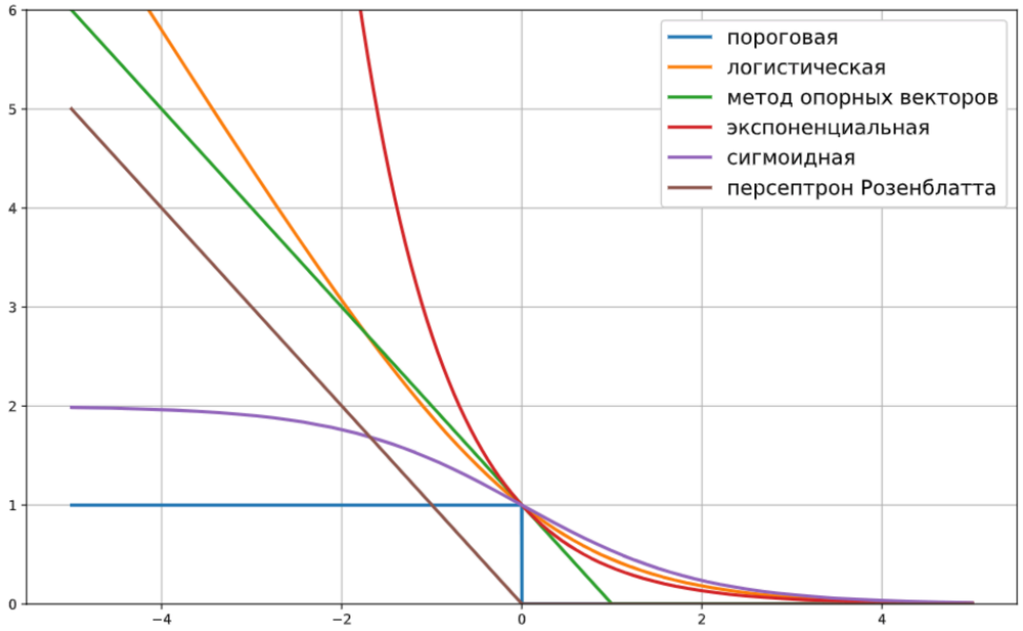


Рис. 1. Верхние оценки на пороговую функцию потерь.

$$Q(a, X) \leqslant \frac{1}{\ell} \sum_{i=1}^{\ell} \tilde{L}(y_i \langle w, x_i \rangle) \to \min_w$$

1. $\tilde{L}(M) = \log\left(1 + e^{-M}\right)$ — логистическая функция потерь
2. $\tilde{L}(M) = (1 - M)_+ = \max(0, 1 - M)$ — кусочно-линейная функция потерь (используется в методе опорных векторов)
3. $\tilde{L}(M) = (-M)_+ = \max(0, -M)$ — кусочно-линейная функция потерь (соответствует персептрону Розенблатта)
4. $\tilde{L}(M) = e^{-M}$ — экспоненциальная функция потерь
5. $\tilde{L}(M) = 2/(1 + e^{M})$ — сигмоидная функция потерь

2. Логистическая регрессия

- Решаем задачу бинарной классификации: $\mathbb{Y} = \{-1, +1\}$

- Минимизация верхней оценки:

$$\frac{1}{\ell}\sum_{i=1}^{\ell}\log(1+\exp(-y_i\langle w, x_i\rangle)) \to \min_{w}$$

2. Логистическая регрессия

Хотим модель, оценивающую вероятность положительного класса:

$$\hat{b}(x) \sim p(y = +1 \,|\, x) \in [0,1]$$

Тогда можно сформулировать следующее требование к функции потерь:

$$Q = \frac{1}{n} \sum_{i=1}^{n} L(y_i, b(x)) \to \min_{b(x)} \Leftrightarrow \text{argmin}_{b(x)} Q \sim p(y = +1 \,|\, x)$$

При устремлении количества экспериментов к бесконечности:

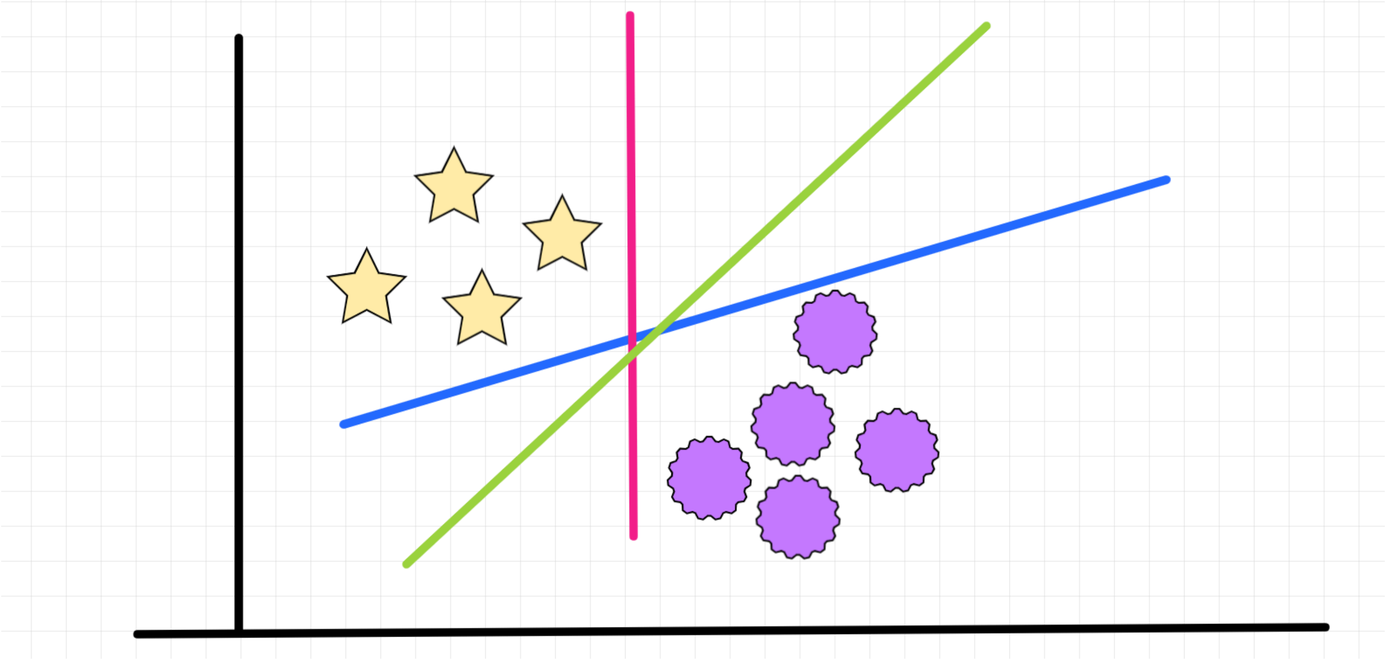
$$n \to \infty : \text{argmin}_{b(x)} \mathbb{E}[L(y, b(x)) \,|\, x] = p(y = +1 \,|\, x)$$

Будем говорить, что модель $b(x)$ предсказывает вероятности, если среди объектов с $b(x) = p$ доля положительных равна p.

SVM:
простое объяснение

3. Общая идея

- *Loss function for SVM*
  - $\mu_i = y_i(\langle w, x \rangle + w_o) \longrightarrow L(\mu) = \max(0,1 - \mu)$
  - *Where μ - margin*



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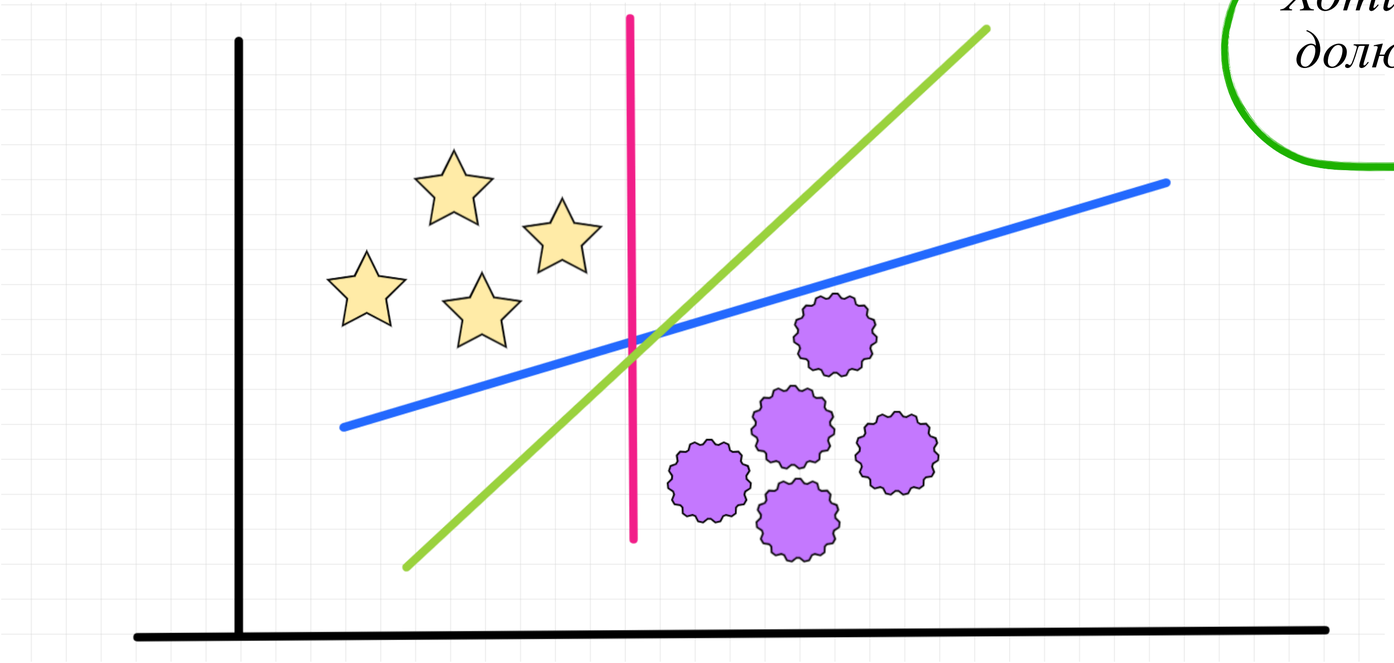
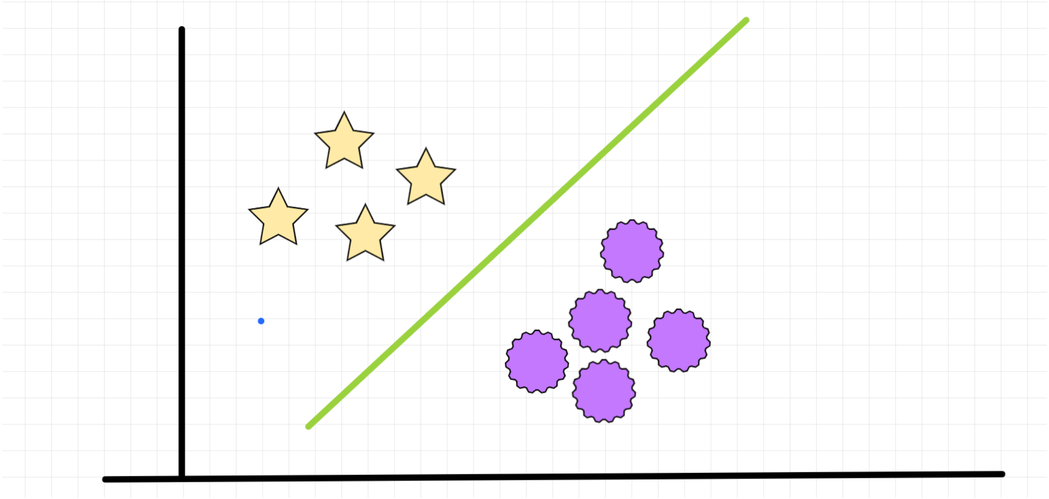


Хотим минимизировать долю предположений о данных

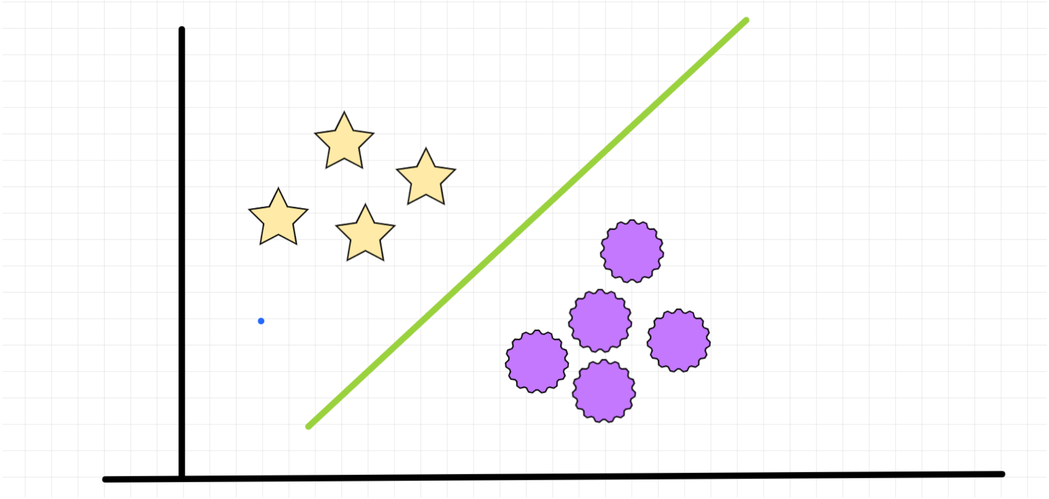
4. Case: divisible



- *Task formulation*
  - $\exists w, w_0 : \; y_i(\langle w, x_i \rangle + w_0) > 0 \; \forall i$
- *First idea*
  - $a(x) = \textbf{sign}(\langle w, x_i \rangle + w_0) = \textbf{sign}(\langle \alpha w, x_i \rangle + \alpha w_0)$
  - *That means we can scale our parameters*
  - *Hence, let's find such values for them that:*

$$\min_{x_i \in \mathbb{X}} |\langle w, x_i \rangle + w_0| = 1$$

4. Case: divisible



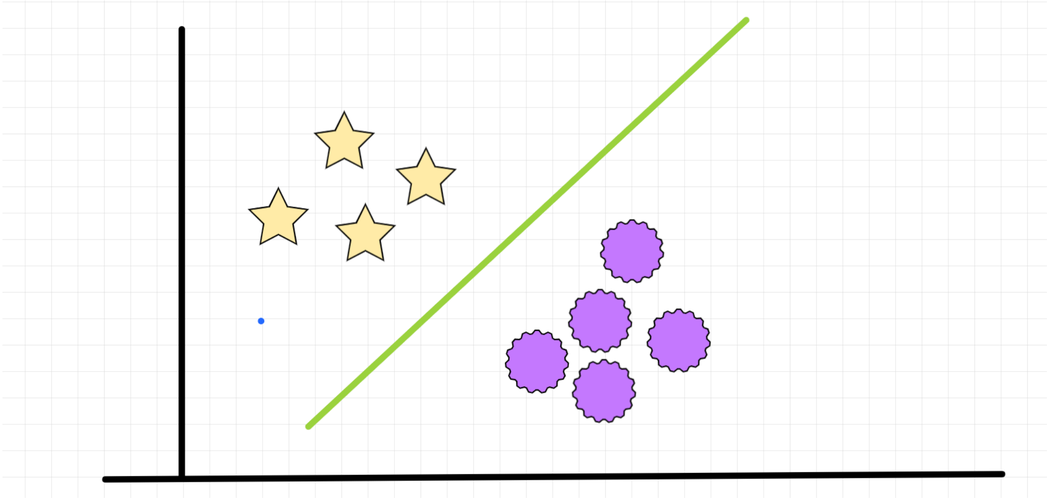
- *First idea*

$$\min_{x_i \in \mathbb{X}} |\langle w, x_i \rangle + w_0| = 1$$

- *Second idea*
  - *We do want to maximise the distance from hyperplane to the closest sample*
  - *Let's evoke the formula from LA for such a distance*

$$\rho(a, x) = \frac{|\langle w, x \rangle + w_0|}{\|w\|} \Longrightarrow \min_{x_i \in \mathbb{X}} \rho(a, x_i) = \min_{x_i \in \mathbb{X}} \frac{|\langle w, x \rangle + w_0|}{\|w\|} = \|w\|^{-1}$$

4. Case: divisible



- *First idea*

$$\min_{x_i \in \mathbb{X}} |\langle w, x_i \rangle + w_0| = 1$$

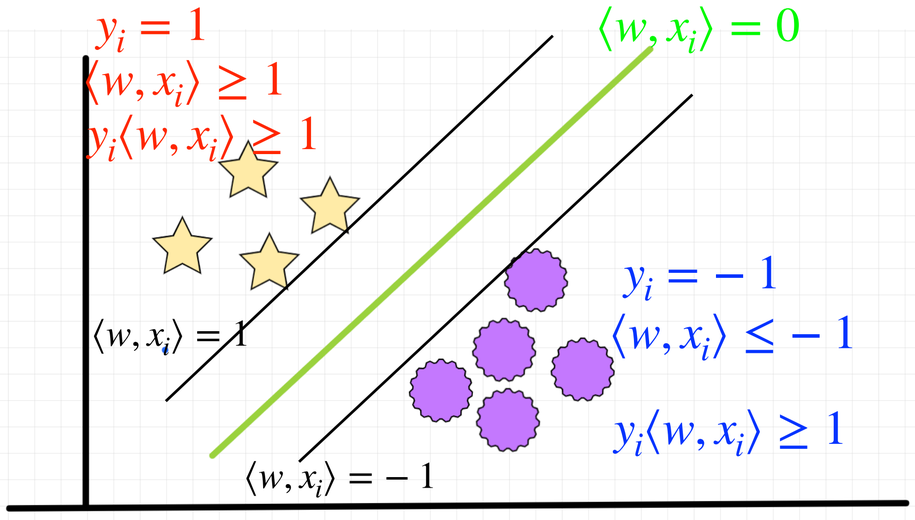
- *Second idea*

$$\rho(a, x) = \frac{|\langle w, x \rangle + w_0|}{\|w\|} \Longrightarrow \min_{x_i \in \mathbb{X}} \rho(a, x_i) = \|w\|^{-1} \min_{x_i \in \mathbb{X}} |\langle w, x \rangle + w_0| = \|w\|^{-1}$$

- *Summarising all together*

$$\begin{cases} \frac{1}{\|w\|} \to \max_w \\ y_i(\langle w, x_i \rangle + w_0) > 0 \ \forall i \\ \min_{x_i \in \mathbb{X}} |\langle w, x_i \rangle + w_0| = 1 \end{cases}$$

4. Case: divisible



- *First idea*

$$\min_{x_i \in \mathbb{X}} |\langle w, x_i \rangle + w_0| = 1$$

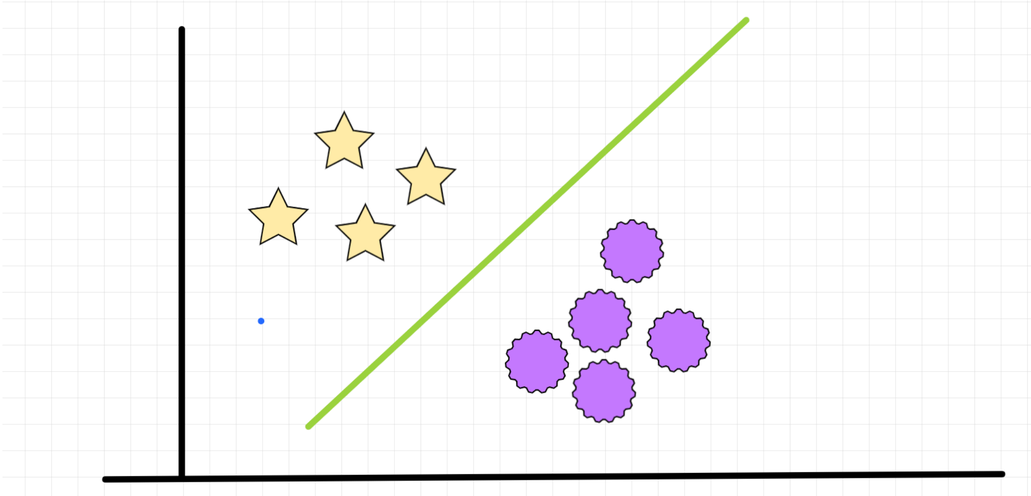
- *Second idea*

$$\rho(a, x) = \frac{|\langle w, x \rangle + w_0|}{\|w\|} \Longrightarrow \min_{x_i \in \mathbb{X}} \rho(a, x_i) = \|w\|^{-1} \min_{x_i \in \mathbb{X}} |\langle w, x \rangle + w_0| = \|w\|^{-1}$$

- *Summarising all together*

$$\begin{cases} \frac{1}{\|w\|} \to \max_{w, w_0} \\ y_i(\langle w, x_i \rangle + w_0) > 0 \ \forall i \\ \min_{x_i \in \mathbb{X}} |\langle w, x_i \rangle + w_0| = 1 \end{cases} \Longrightarrow \begin{cases} \|w\|^2 \to \min_{w, w_0} \\ y_i(\langle w, x_i \rangle + w_0) \geq 1 \end{cases}$$

4. Case: divisible



- *First idea*

$$\min_{x_i \in \mathbb{X}} |\langle w, x_i \rangle + w_0| = 1$$

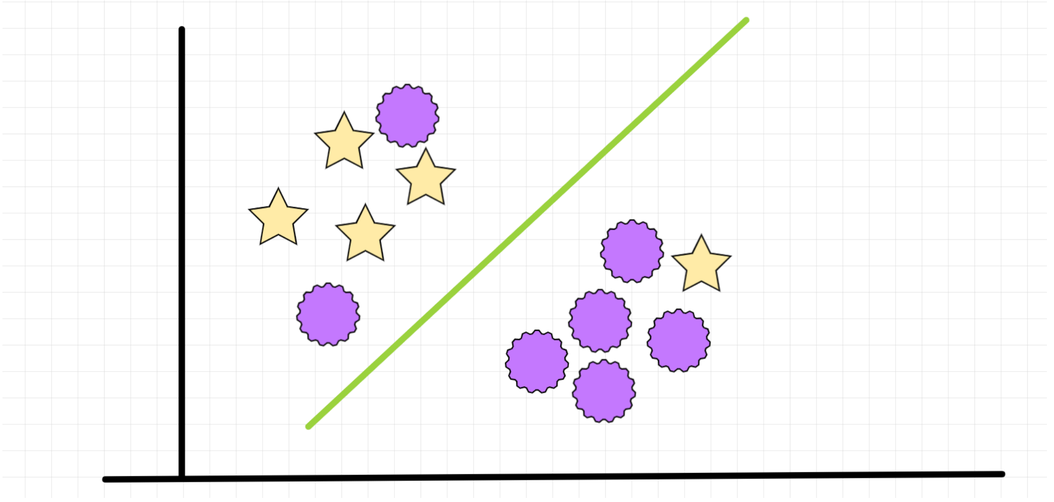
- *Second idea*

$$\rho(a, x) = \frac{|\langle w, x \rangle + w_0|}{\|w\|} \Longrightarrow \min_{x_i \in \mathbb{X}} \rho(a, x_i) = \|w\|^{-1} \min_{x_i \in \mathbb{X}} |\langle w, x \rangle + w_0| = \|w\|^{-1}$$

- *Summarising all together*

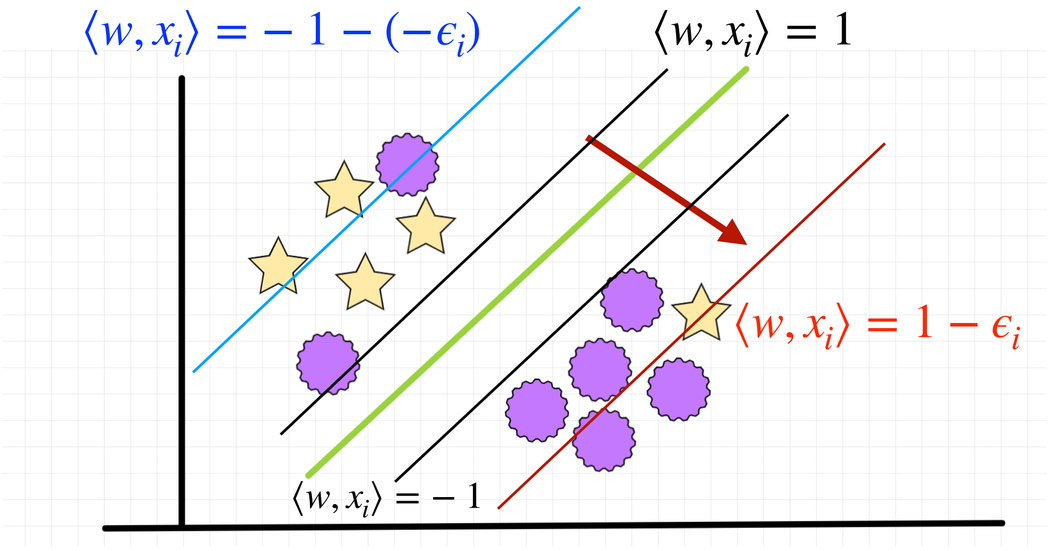
$$\begin{cases} \|w\|^{-1} \to \max_{w, w_0} \\ y_i(\langle w, x_i \rangle + w_0) > 0 \ \forall i \\ \min_{x_i \in \mathbb{X}} |\langle w, x_i \rangle + w_0| = 1 \end{cases} \Longrightarrow \begin{cases} \|w\|^2 \to \min_{w, w_0} \\ y_i(\langle w, x_i \rangle + w_0) \geq 1 \end{cases}$$

5. Case: ***in***divisible



- *Task formulation*
  - $\forall w, w_0 : \ \exists x_i \in \mathbb{X} : \ y_i(\langle w, x_i \rangle + w_0) < 0$
- *First idea*
  - *We still want to maximise the distance to the closest object by minimising weights' norm*
- *Second idea*
  - *Our condition for margin to be greater than 1 for all objects is too strict*
  - *Let's ease it, but also try to minimise the extent of relaxation*

5. Case: *in*divisible



- *First idea*
  - *We still want to maximise the distance to the closest object by minimising weights' norm*
- *Second idea*
  - *Our condition for margin to be greater than 1 for all objects is too strict*
  - *Let's ease it, but also try to minimise the extent of relaxation*
- *Summarising all together*

$$\begin{cases} \frac{1}{2}\|w\|^2 \to \min_{w,w_0} \\ y_i(\langle w, x_i \rangle + w_0) \geq 1 - \epsilon_i \end{cases} \Longrightarrow \begin{cases} \frac{1}{2}\|w\|^2 + C\sum \epsilon_i \to \min_{w,w_0} \\ y_i(\langle w, x_i \rangle + w_0) \geq 1 - \epsilon_i \\ \epsilon_i \geq 0 \end{cases}$$

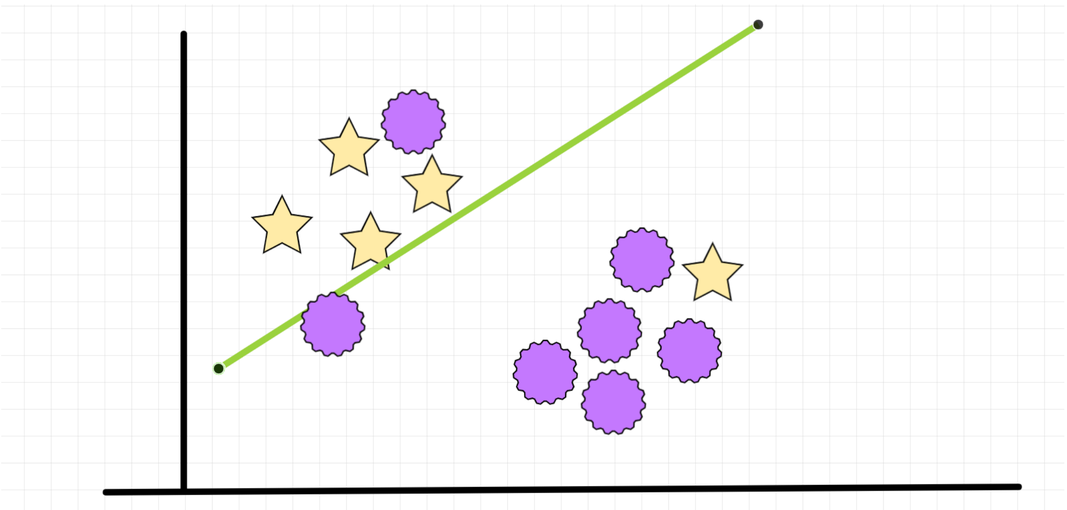
5. Case: *in*divisible

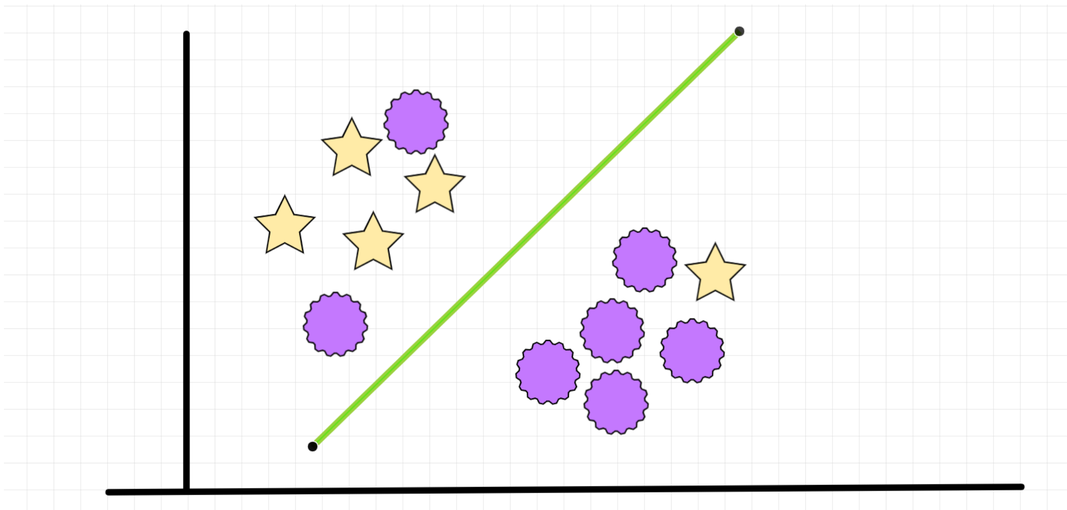
- *Summarising all together*

Penalty for a small distance to the closest object

Penalty for model's errors

$$\begin{cases} \frac{1}{2}\|w\|^2 + C\sum \epsilon_i \to \min_{w,w_0} \\ y_i(\langle w, x_i\rangle + w_0) \geq 1 - \epsilon_i \\ \epsilon_i \geq 0 \end{cases}$$





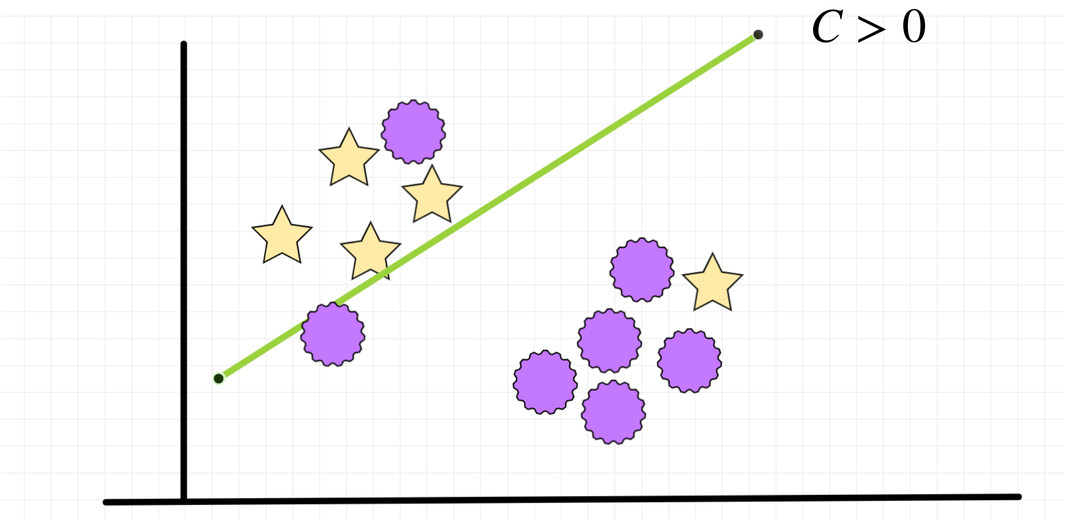
5. Case: *in*divisible

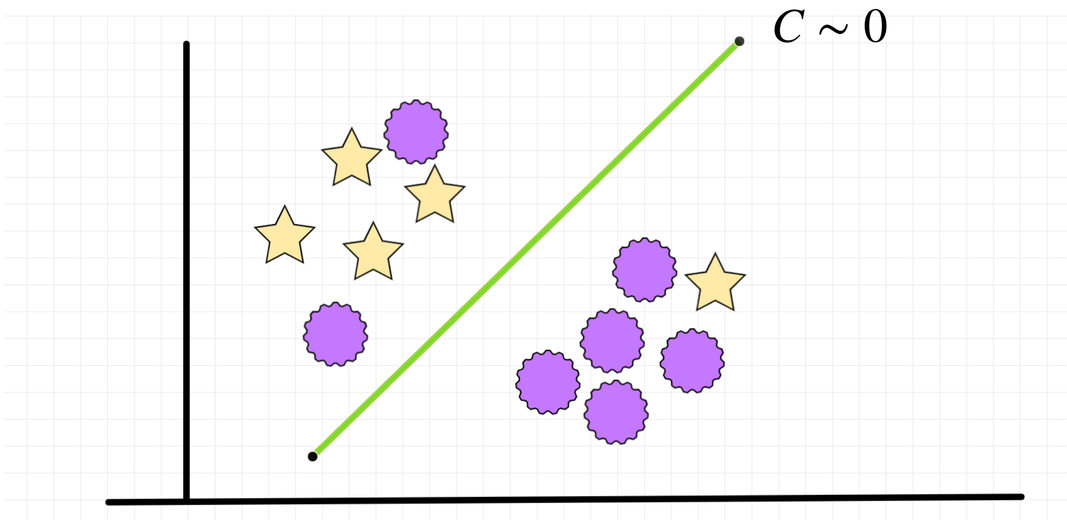
- *Summarising all together*

Penalty for a small distance to the closest object

Penalty for model's errors

$$\begin{cases} \frac{1}{2}\|w\|^2 + C\sum \epsilon_i \to \min_{w,w_0} \\ y_i(\langle w, x_i\rangle + w_0) \geq 1 - \epsilon_i \\ \epsilon_i \geq 0 \end{cases}$$





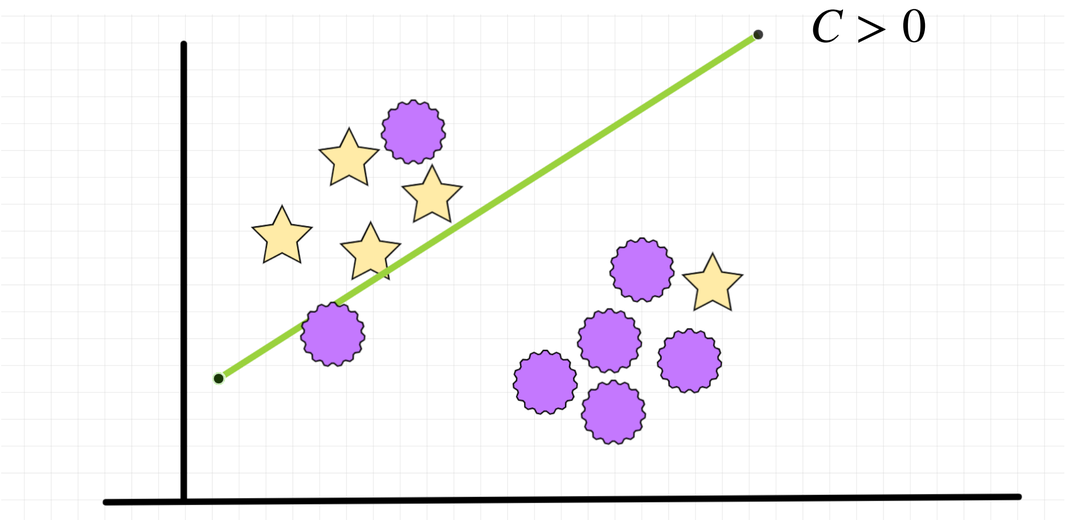
5. Case: *in*divisible

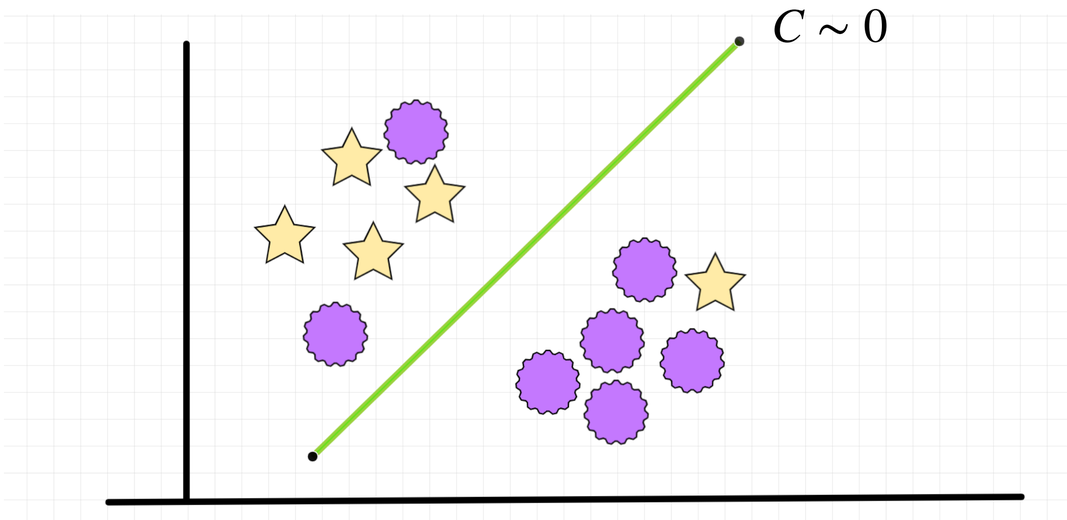
- *Summarising all together*

Penalty for a small distance to the closest object

***Why** L1, not L2?*

$$\begin{cases} \frac{1}{2}\|w\|^2 + C\sum \epsilon_i \to \min_{w,w_0} \\ y_i(\langle w, x_i\rangle + w_0) \geq 1 - \epsilon_i \\ \epsilon_i \geq 0 \end{cases}$$

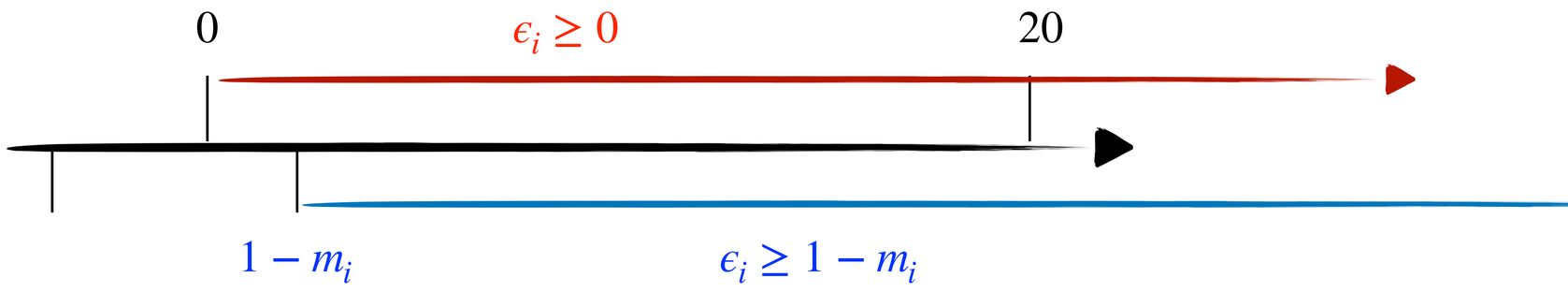




5. Case: *in*divisible

- *Reducing № of conditions*

$$\begin{cases} \frac{1}{2}\|w\|^2 + C\sum \epsilon_i \to \min_{w,w_0} \\ \epsilon_i \geq 1 - y_i(\langle w, x_i\rangle + w_0) \\ \epsilon_i \geq 0 \end{cases} \xRightarrow{\epsilon \to \min} \begin{cases} \frac{1}{2}\|w\|^2 + C\sum \epsilon_i \to \min_{w,w_0} \\ \epsilon_i = \max(0,\ 1 - y_i(\langle w, x_i\rangle + w_0)) \end{cases}$$



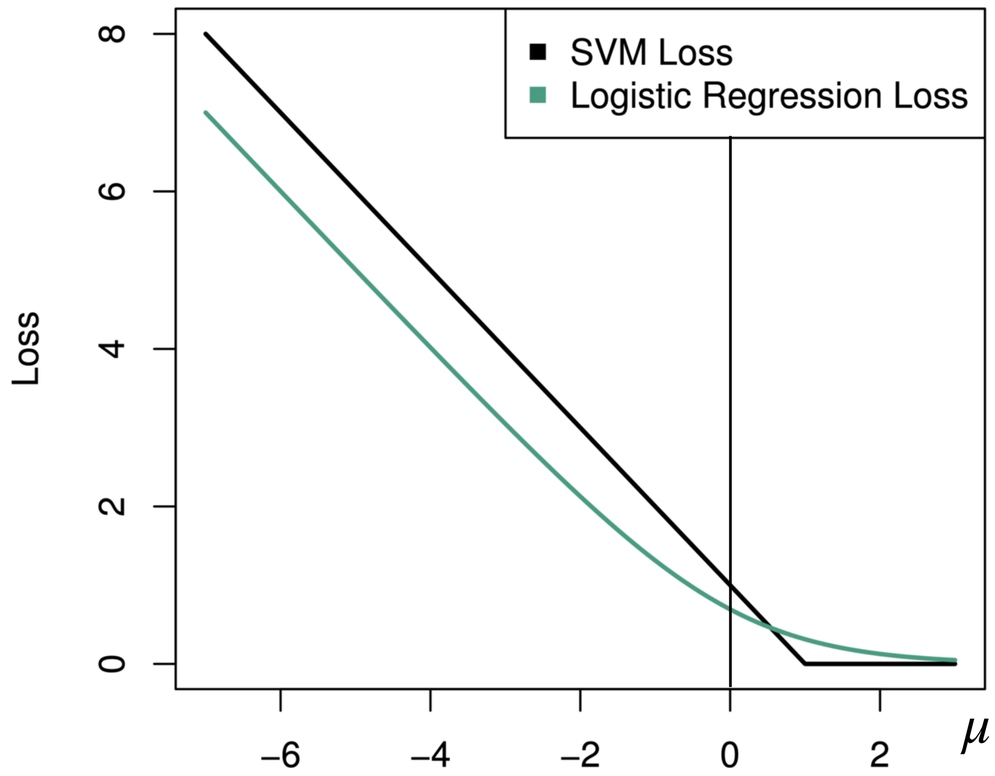
5. Case: *in*divisible

- *Reducing № of conditions*

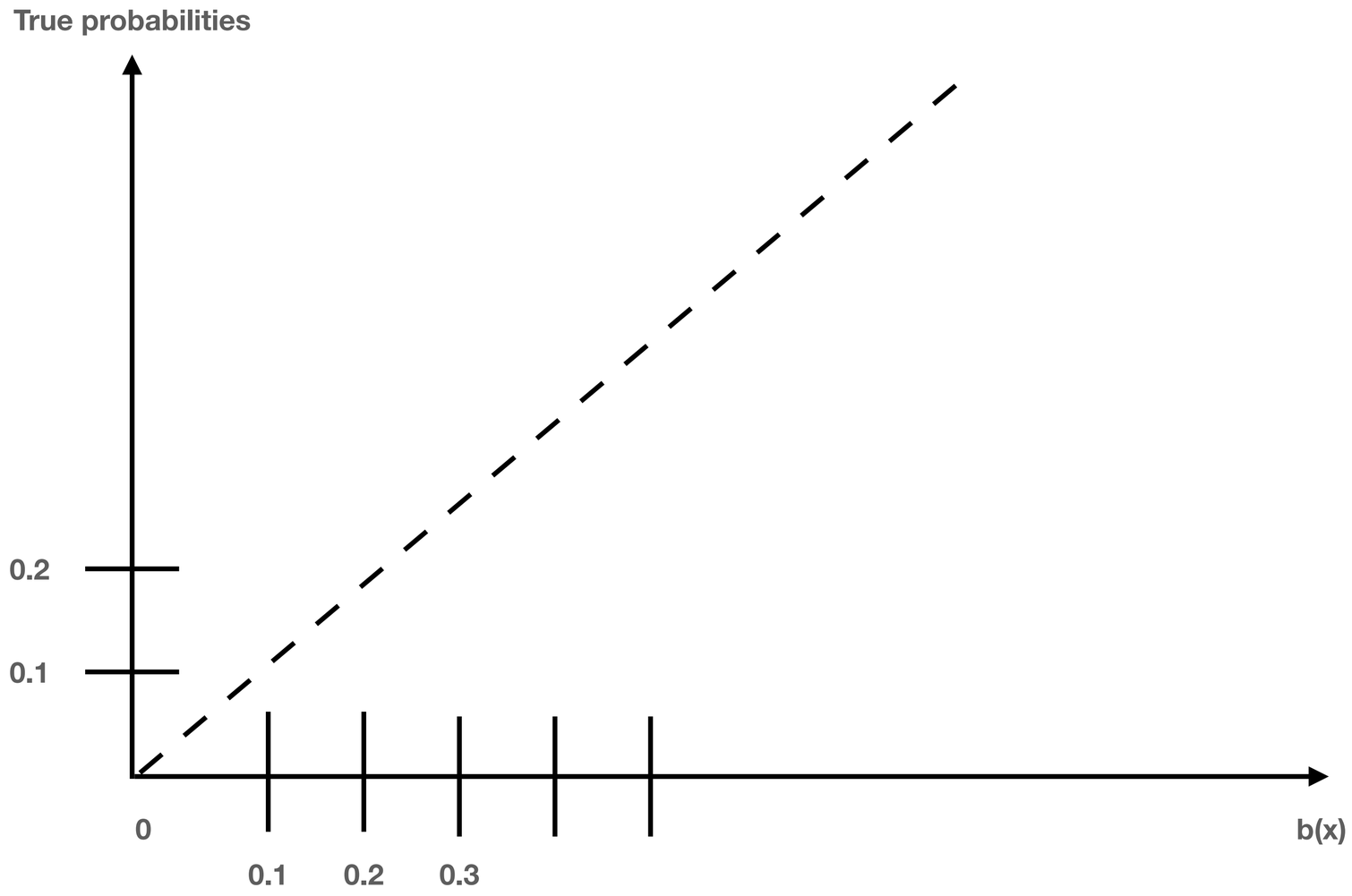
$$\begin{cases} \frac{1}{2}\|w\|^2 + C\sum \epsilon_i \to \min_{w,w_0} \\ \epsilon_i \geq 1 - y_i(\langle w, x_i\rangle + w_0) \\ \epsilon_i \geq 0 \end{cases} \overset{\epsilon\to\min}{\Longrightarrow} \begin{cases} \frac{1}{2}\|w\|^2 + C\sum \epsilon_i \to \min_{w,w_0} \\ \epsilon_i = \max(0,1 - y_i(\langle w, x_i\rangle + w_0)) \end{cases}$$

$$\Longrightarrow \underbrace{\frac{1}{2}\|w\|^2}_{\textit{L2 regularisation}} + \underbrace{C\sum \max(0,1 - y_i(\langle w, x_i\rangle + w_0))}_{\textit{Loss functional}} \to \min_{w,w_0}$$

6. Compare to *LR*



- *In terms of μ:*
  - $\mu < 0$: *act identically*
  - $\mu > 0$: ***SVM*** *goes to zero as μ approaches 1, while* ***LR*** *forces μ to raise till infinity*
- *Overall Inference:*
  - ***SVM*** *on average is better for classification metrics*
  - ***LR*** *struggles to be more accurate at estimating probabilities*


True probabilities
0.2
0.1
0
0.1
0.2
0.3
b(x)